

12.684

EE25BTECH11047 - RAVULA SHASHANK REDDY

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Question:

If $\mathbf{A}$ is a square matrix then the orthogonality property mandates:

- a) $\mathbf{A}\mathbf{A}^\top = \mathbf{I}$ b) $\mathbf{A}\mathbf{A}^\top = 0$ c) $\mathbf{A}\mathbf{A}^\top = \mathbf{A}^{-1}$ d) $\mathbf{A}\mathbf{A}^\top = \mathbf{A}$

Solution:

A square matrix $\mathbf{A}$ is orthogonal if its transpose equals its inverse:

$$\mathbf{A}^\top = \mathbf{A}^{-1} \tag{1}$$

Multiplying both sides on the left by $\mathbf{A}$:

$$\begin{aligned} \mathbf{A}\mathbf{A}^\top &= \mathbf{A}\mathbf{A}^{-1} \\ \Rightarrow \mathbf{A}\mathbf{A}^\top &= \mathbf{I} \end{aligned} \tag{2}$$